Solutions: Day 33 Practice Problems Suite

Solution to Case Study 1: Feature Transformation & RFM Pipeline Design

1. Mathematical Flaw of Raw Timestamps & Birth Years

In Euclidean distance spaces:

$$d(u, v) = \sqrt{\sum_{i=1}^{p} (u_i - v_i)^2}$$

1. **Raw Birth Year Scale Distortion**: Birth years (e.g., 1960 vs 1990) have squared differences $(1990 - 1960)^2 = 900$. However, birth year is an arbitrary origin-based interval metric. Translating it to `Age` $(2026 - \text{Year_Birth})$ grounds the feature in direct life-cycle relevance.
2. **Raw Date String Incomputability**: Date strings (`"2021-08-14"`) cannot be parsed by inner products. If parsed as raw POSIX timestamps (1.62×10^9 seconds), the enormous numerical magnitude completely dominates all other normalized features. Transforming dates into `Customer_Tenure_Days` yields an interpretable measure of loyalty duration.

2. Reproducibility Anchor Date

- **Pitfall**: Using dynamic current system time `datetime.now()` causes the tenure of every customer to increase dynamically every time the pipeline executes.
- **Solution**: Anchor the reference date strictly to the dataset maximum:

$$\text{Reference Date} = \max(\text{Dt_Customer})$$

$$\text{Customer_Tenure_Days}_i = (\text{Reference Date} - \text{Dt_Customer}_i)$$

This guarantees that running the pipeline on future compute clusters yields identical vector coordinates and identical cluster centroids.

3. The `drop_first=True` Fallacy in Clustering

- In OLS regression, the parameter vector is solved via normal equations:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

If all dummy columns are retained, $\sum_{j=1}^{C} D_j = \mathbf{1}$, causing $X^T X$ to be singular (non-invertible). Dropping the first dummy avoids this singularity.

- In unsupervised clustering, no matrix inversion occurs. Clustering measures the geometric distance between points:

$$d(D_A, D_B) = \sqrt{(1-0)^2 + (0-1)^2} = \sqrt{2}$$

If category 1 is dropped, points in category 1 become $(0, 0, \ldots, 0)$, while categories 2 and 3 remain unit vectors. Consequently, the distance between category 1 and category 2 is:

$$d(D_1, D_2) = \sqrt{(0-1)^2 + 0} = 1 \neq \sqrt{2}$$

Dropping the first column destroys geometric symmetry. Therefore, in clustering, **all categorical levels must be retained** (`drop=None`).

Solution to Case Study 2: Quantitative Hyperparameter Tuning

1. WCSS Formulation & Monotonicity

Within-Cluster Sum of Squares (Inertia) is defined as:

$$\text{WCSS}(K) = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

- When $K = 1$, all data points are measured against the global centroid μ.
- When K increases to $K + 1$, splitting any cluster into two sub-centroids μ_{k1}, μ_{k2} cannot increase variance because:

$$\min_{\mu_1,\mu_2}\left(\sum_{x\in C_{k1}}\|x-\mu_1\|^2+\sum_{x\in C_{k2}}\|x-\mu_2\|^2\right)\leq\sum_{x\in C_k}\|x-\mu_k\|^2$$

- In the extreme limit where $K=N$, every data point becomes its own centroid ($x_i=\mu_i$), resulting in $\text{WCSS}=0$. Hence, WCSS decreases monotonically with K.

2. `KneeLocator` Mathematical Formulation

The `KneeLocator` algorithm normalizes both K and WCSS onto the unit interval $[0,1]$:

$$x_{\text{norm}}(k)=\frac{k-k_{\min}}{k_{\max}-k_{\min}},\quad y_{\text{norm}}(k)=\frac{\text{WCSS}(k)-\text{WCSS}_{\min}}{\text{WCSS}_{\max}-\text{WCSS}_{\min}}$$

It constructs a diagonal reference chord between the first point $(0,1)$ and last point $(1,0)$:

$$\text{Chord}(k)=1-x_{\text{norm}}(k)$$

The knee is the index k^* that maximizes the orthogonal perpendicular distance from the curve to the chord:

$$k^*=\arg\max_k\left(\text{Chord}(k)-y_{\text{norm}}(k)\right)$$

3. Metric Evaluation Table & Optimal Selection

K	WCSS	$\Delta\text{WCSS}=\text{WCSS}_{K-1}-\text{WCSS}_K$	Silhouette Score
2	24,500	—	0.38
3	16,800	7,700	0.42
4	**11,200**	**5,600**	**0.45 (Maximum)**
5	9,900	1,300 (Sharp deceleration)	0.41
6	8,900	1,000	0.39
7	8,100	800	0.36

K	WCSS	$\Delta\text{WCSS} = \text{WCSS}_{K-1} - \text{WCSS}_K$	Silhouette Score
8	7,400	700	0.34

- **Marginal Drop Analysis**: Moving from $K = 3 \to 4$ reduces WCSS by 5,600 units. Moving from $K = 4 \to 5$ reduces WCSS by only 1,300 units (a 76.8% deceleration in variance reduction).
- **Silhouette Confirmation**: The global silhouette score peaks precisely at $K = 4$ ($s = 0.45$).
- **Conclusion**: Both the Elbow Method and Silhouette Analysis unambiguously confirm $K^* = 4$.

Solution to Case Study 3: Cluster Persona Strategy & Marketing ROI Optimization

1. Persona Mapping

1. **Cluster Alpha**: *"Family Shoppers / Thrifty Couples"* — Moderate income (\$38.5k), modest spend (\$210), high dependents (1.35), high browsing (6.8 visits), low responsiveness (6.5%).
2. **Cluster Beta**: *"Affluent Couples / Established Spenders"* — High income (\$74.2k), highest spending (\$1,280), low dependents (0.42), low browsing (3.4 visits), solid responsiveness (17.2%).
3. **Cluster Gamma**: *"Budget Singles / Window Shoppers"* — Lowest income (\$35.8k), lowest spend (\$155), single parents (1.40 kids), highest web visits (7.1), moderate response (13.8%).
4. **Cluster Delta**: *"Affluent Singles / Golden Cluster"* — High income (\$71.9k), high spend (\$1,210), minimal dependents (0.38), extraordinary response rate (**33.5%**).

2. Quantitative Justification of Delta as the "Golden Cluster"

Consider expected revenue generated per marketing contact:

- Expected Revenue per Contact:

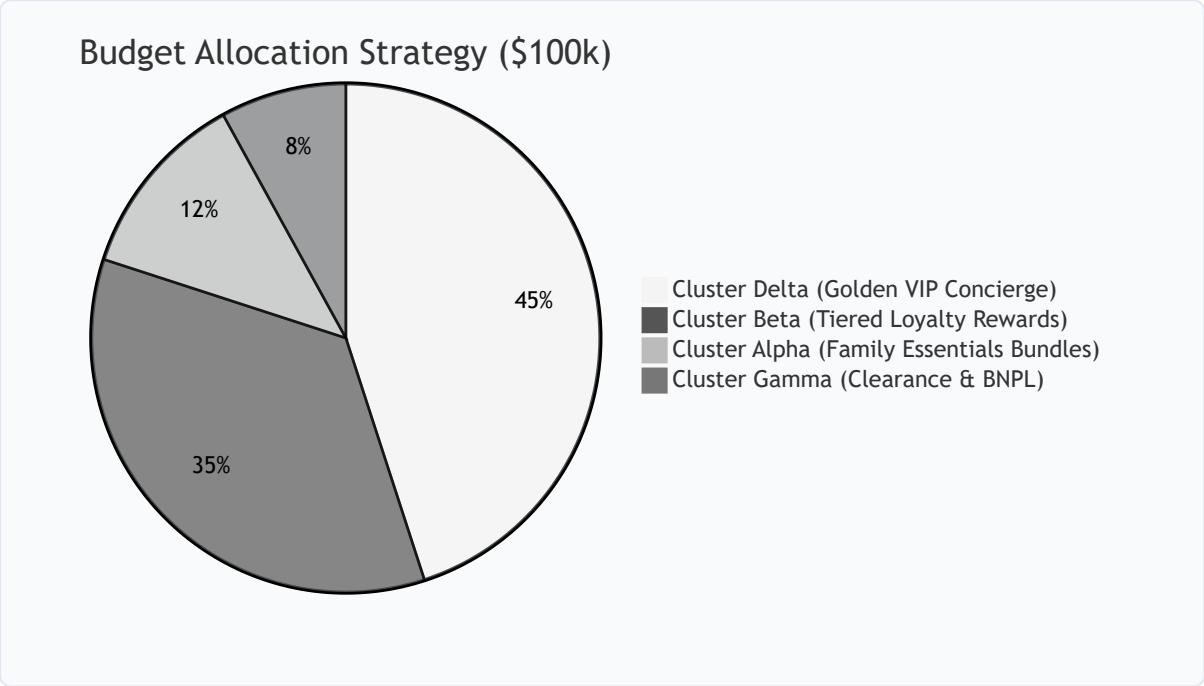
$$E[\text{Revenue}] = \text{Response Rate} \times \text{Mean Spend}$$

- **Cluster Alpha**: \$0.065 \times \$210 = \mathbf{\13.65} per customer contacted.
- **Cluster Beta**: \$0.172 \times \$1,280 = \mathbf{\220.16} per customer contacted.

- **Cluster Gamma**: $0.138 \times \$155 = \mathbf{\$21.39}$ per customer contacted.
- **Cluster Delta**: $0.335 \times \$1,210 = \mathbf{\$405.35}$ per customer contacted!

Finding: Reaching out to a Cluster Delta customer yields **$405.35 in expected return**, which is **1.84x higher than Beta** and **nearly 30x higher than Alpha**! This exceptional capital efficiency establishes Delta as the undisputed Golden Cluster.

3. Marketing Strategy & Budget Allocation ($100,000 Total Budget)



1. **Cluster Delta (Allocation: $45,000 / 45%)**:
    - High-touch 1-on-1 concierge marketing, personalized gourmet/luxury recommendations, private preview events.
2. **Cluster Beta (Allocation: $35,000 / 35%)**:
    - VIP tiered loyalty program with milestone tier gifts and luxury physical catalog mailers.
3. **Cluster Alpha (Allocation: $12,000 / 12%)**:
    - Automated email coupon campaigns, weekend family discount bundles, back-to-school essentials.
4. **Cluster Gamma (Allocation: $8,000 / 8%)**:
    - Automated push notifications for clearance flash sales, abandoned cart recovery coupons, and BNPL installment promotions.